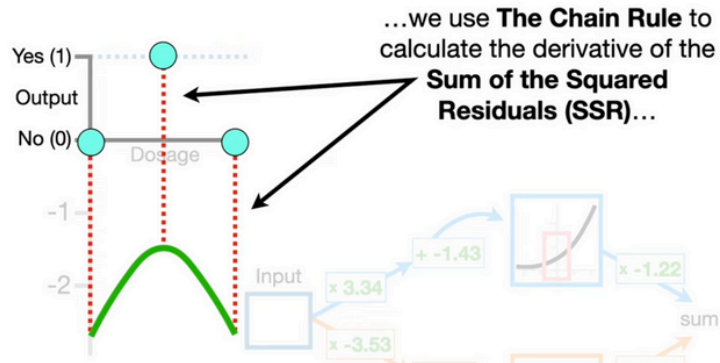
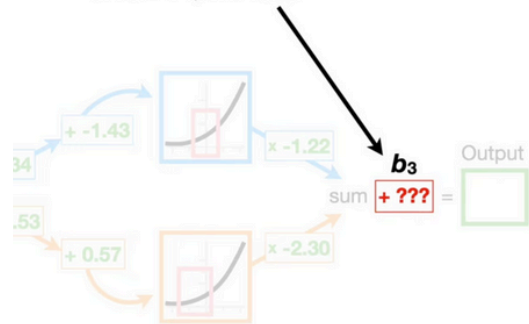


Backpropagation

Main idea

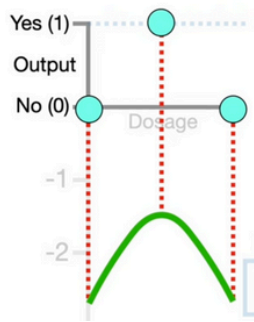
So the **Main Ideas** for **Backpropagation** are that when a parameter is unknown, like b_3 ...



...we use **The Chain Rule** to calculate the derivative of the **Sum of the Squared Residuals (SSR)**...



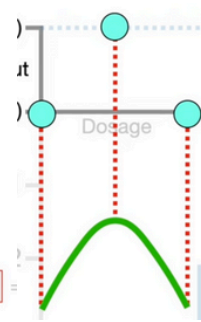
$$\frac{d SSR}{d b_3} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d b_3}$$



...with respect to the unknown parameter, which in this case was b_3 .



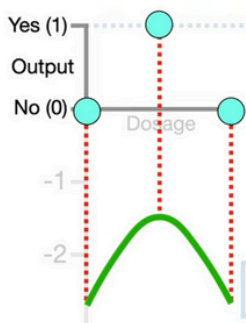
$$\frac{d SSR}{d b_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times 1$$



...with respect to the unknown parameter, which in this case was b_3 .



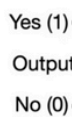
$$\frac{d SSR}{d b_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times 1$$



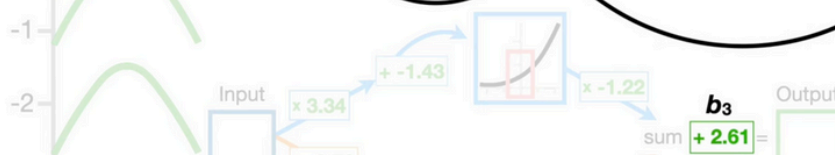
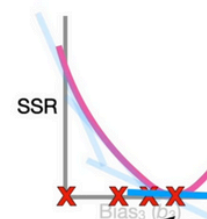
Then we initialize the unknown parameter with a number, and in this case we set $b_3 = 0$...



$$\frac{d SSR}{d b_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times 1$$

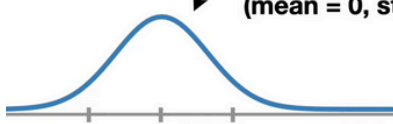


...and used **Gradient Descent** to optimize the unknown parameter.

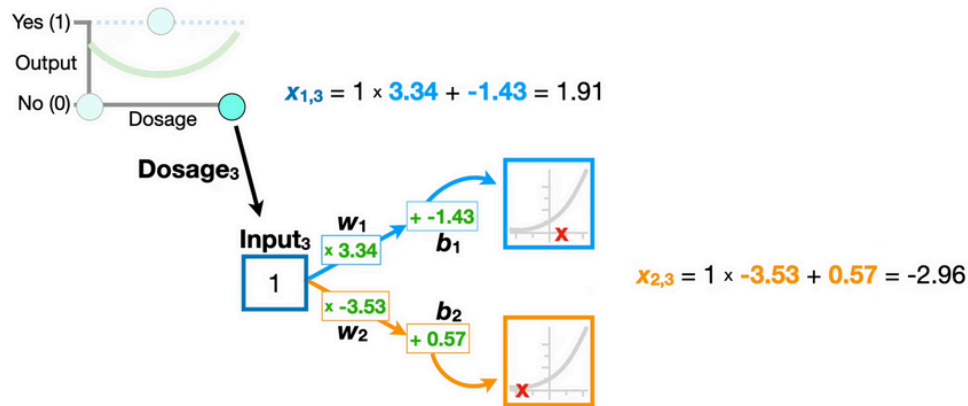
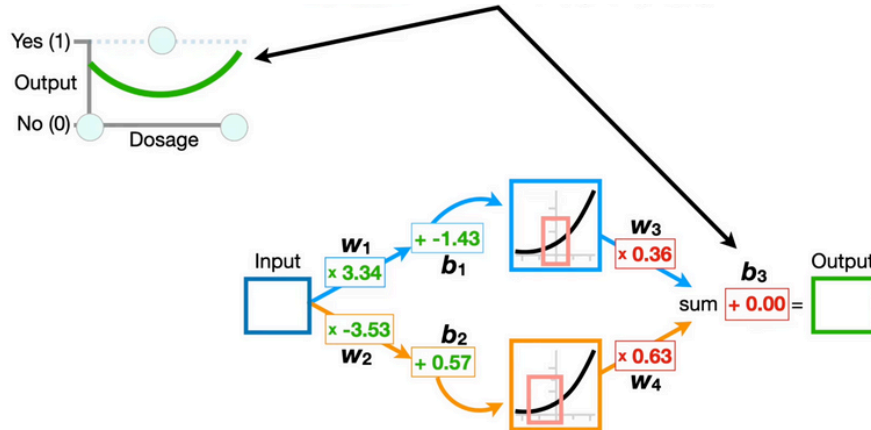


BPP Details

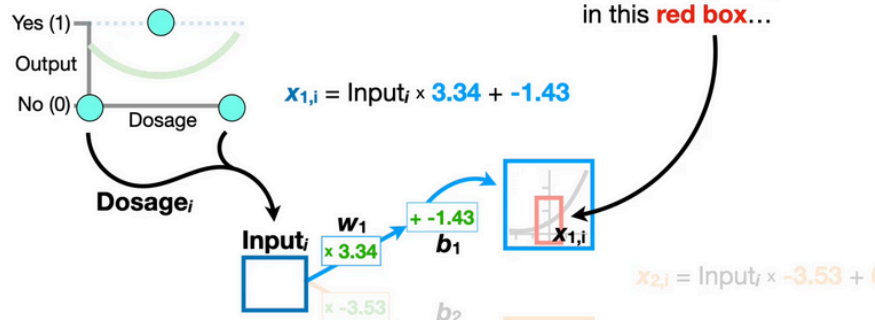
...and, in this example, that means we randomly select 2 values from a **Standard Normal Distribution** (mean = 0, standard deviation = 1).



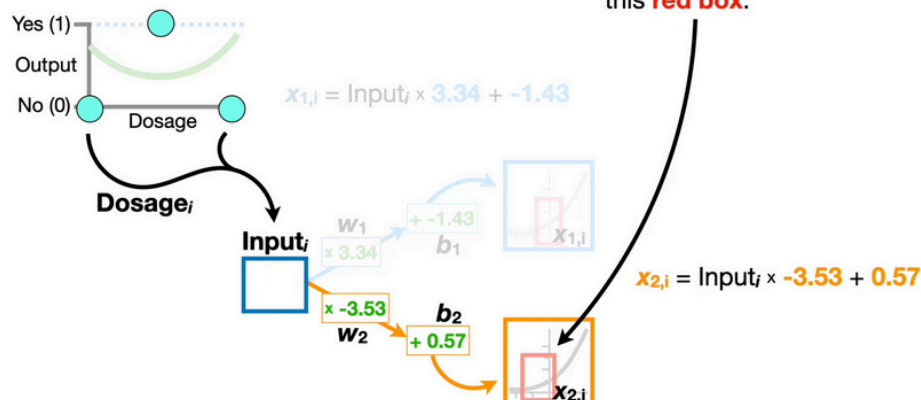
Then we initialize the last **Bias**, b_3 , to 0, because **Bias** terms frequently start at 0.



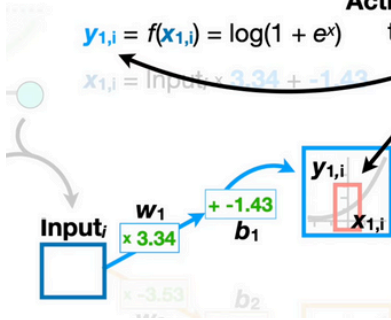
If we plugged in all values for i into **Dosage_i**, we get $x_{1,i}$ values in this **red box**...



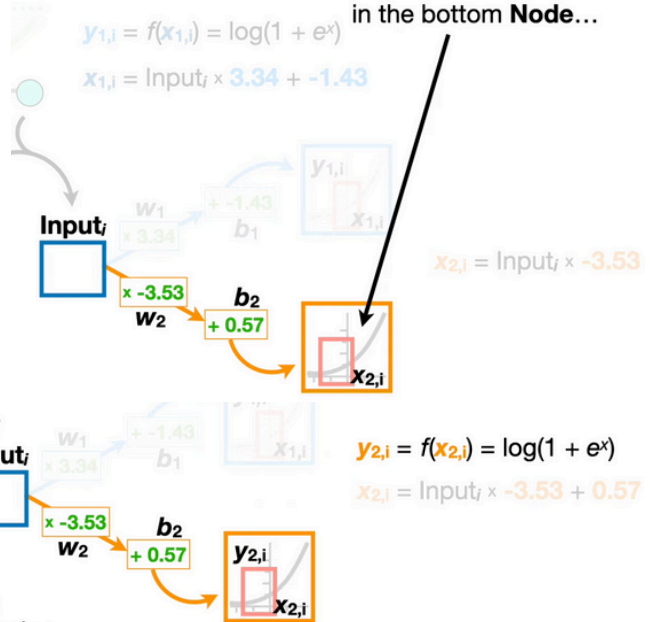
...and $x_{2,i}$ values in this **red box**.



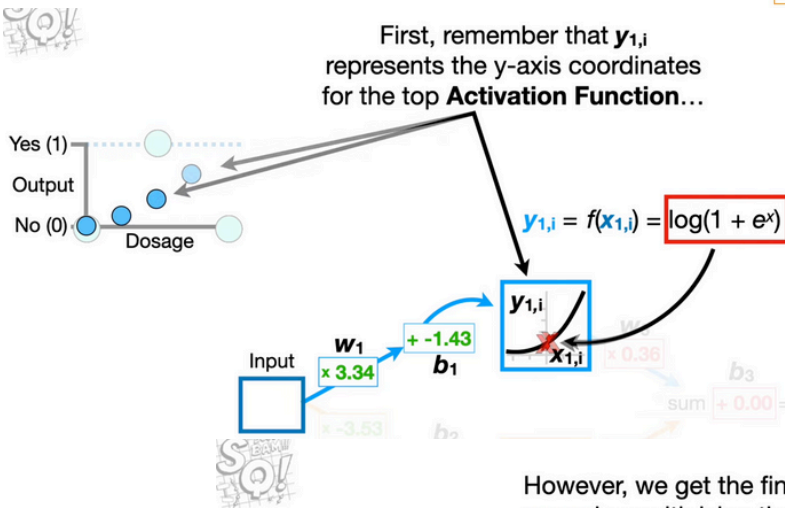
Just like before, the 1 in $1,i$ tells us that we are talking about the **Activation Function** in the top **Node**...



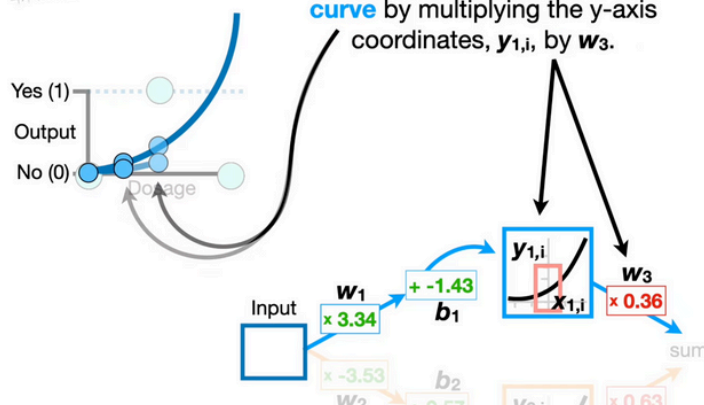
Likewise, in order to get the y-axis coordinates for the **Activation Function** in the bottom **Node**...



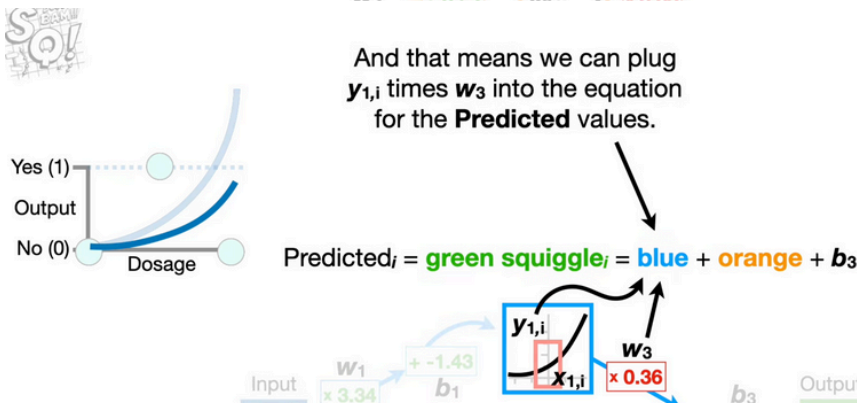
First, remember that $y_{1,i}$ represents the y-axis coordinates for the top **Activation Function**...



However, we get the final **blue curve** by multiplying the y-axis coordinates, $y_{1,i}$, by w_3 .

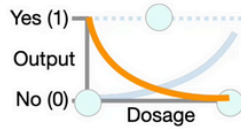


And that means we can plug $y_{1,i}$ times w_3 into the equation for the **Predicted** values.

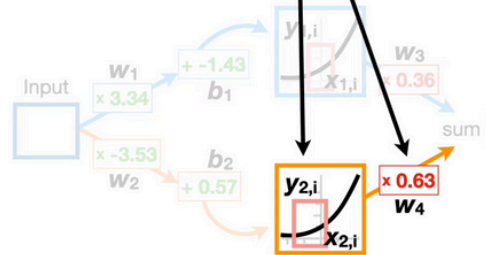




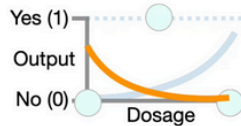
Likewise w_4 multiplies the y-axis coordinates, $y_{2,i}$, from the bottom **Activation Function**...



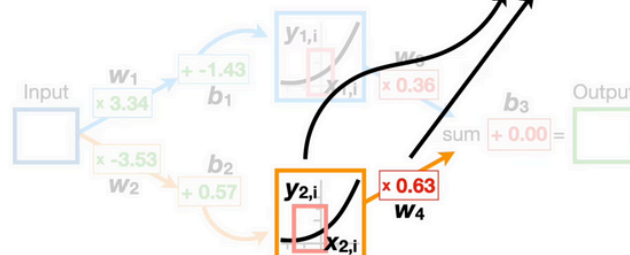
$$\text{Predicted}_i = \text{green squiggle}_i = y_{1,i}w_3 +$$



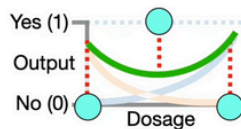
And that means we can plug $y_{2,i}$ times w_4 into the equation for the **Predicted** values.



$$\text{Predicted}_i = \text{green squiggle}_i = y_{1,i}w_3 + y_{2,i}w_4 + b_3$$



...that we evaluate with the **SSR**...

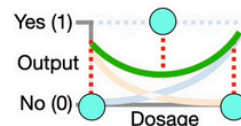


$$\text{SSR} = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{green squiggle}_i = y_{1,i}w_3 + y_{2,i}w_4 + b_3$$



...then the **SSR** are linked to w_3 and w_4 ...



$$\text{SSR} = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{green squiggle}_i = y_{1,i}w_3 + y_{2,i}w_4 + b_3$$

$$\frac{d \text{SSR}}{d w_3}$$

...to determine the derivative of the **SSR** with respect to w_3 ...

$$\text{SSR} = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{green squiggle}_i = y_{1,i}w_3 + y_{2,i}w_4 + b_3$$

$$\frac{d SSR}{d w_3} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_3}$$

...times the derivative of the **Predicted** values with respect to w_3 .

$$SSR = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{green squiggle}_i = y_{1,i} w_3 + y_{2,i} w_4 + b_3$$

$$\frac{d SSR}{d w_4} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_4}$$

...times the derivative of the **Predicted** values with respect to w_4 .

$$SSR = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{green squiggle}_i = y_{1,i} w_3 + y_{2,i} w_4 + b_3$$

So!

$$\frac{d SSR}{d w_3} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_3}$$

$$\frac{d SSR}{d w_4} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_4}$$

...then we use The Chain Rule...

$$\frac{d SSR}{d \text{ Predicted}} = \frac{d}{d \text{ Predicted}} \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

So!

$$\frac{d SSR}{d w_3} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_3}$$

$$\frac{d SSR}{d w_4} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_4}$$

...and this is the derivative of the **SSR** with respect to the **Predicted** values.

$$\frac{d SSR}{d \text{ Predicted}} = \frac{d}{d \text{ Predicted}} \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2 = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i)$$

So!

$$\frac{d SSR}{d w_3} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i)$$

$$\frac{d SSR}{d w_4} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_4} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i)$$

...we plug in the equation for the **Predicted** values.

$$\frac{d \text{ Predicted}}{d w_3} = \frac{d}{d w_3} \text{green squiggle} = \frac{d}{d w_3} (y_{1,i} w_3 + y_{2,i} w_4 + b_3)$$

$$SSR = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{green squiggle}_i = y_{1,i} w_3 + y_{2,i} w_4 + b_3$$



$$\frac{d SSR}{d w_3} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i)$$

$$\frac{d SSR}{d w_4} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_4} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times y_{2,i}$$

...and the derivatives of the other terms are both 0 since they do not contain w_3 .

$$\frac{d \text{ Predicted}}{d w_3} = \frac{d}{d w_3} \text{ green squiggle} = \frac{d}{d w_3} (y_{1,i}w_3 + y_{2,i}w_4 + b_3) = y_{1,i} + 0 + 0$$

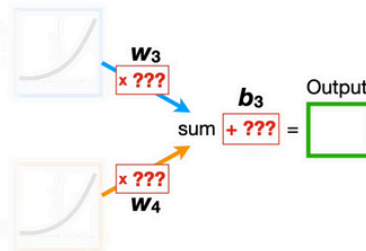


$$\frac{d SSR}{d w_3} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times y_{1,i}$$

$$\frac{d SSR}{d w_4} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d w_4} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times y_{2,i}$$

$$\frac{d SSR}{d b_3} = \frac{d SSR}{d \text{ Predicted}} \times \frac{d \text{ Predicted}}{d b_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times 1$$

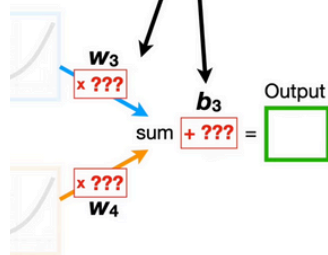
...we can plug them into **Gradient Descent** to optimize w_3 , w_4 and b_3 .



$y_{1,i} \times y_{2,i}$

First, we initialize w_3 and w_4 with random values and set $b_3 = 0$.

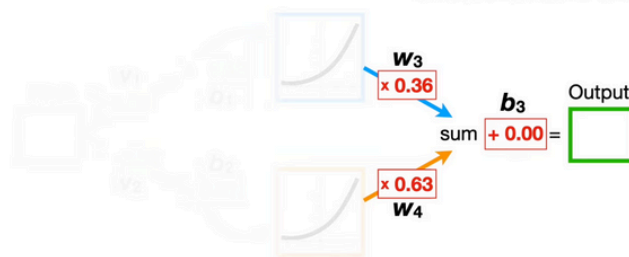
$y_{1,i} \times 1$



$$\frac{d SSR}{d w_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times y_{1,i}$$

Now, starting with the derivative of the **SSR** with respect to w_3 ...

(NOTE: It does not matter which derivative we start with.)



$$\frac{d SSR}{d w_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times y_{1,i}$$

$$\frac{d SSR}{d w_4} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times y_{2,i}$$

$$\frac{d SSR}{d b_3} = \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times 1$$

Now we repeat that process until the **Predictions** no longer improve very much, or we reach a maximum number of steps or we meet some other criteria.

